

Warfarin

1. Vitamin K medic stopping the bleeding: vitamin K is a cofactor for the enzymatic activation of clotting factors
2. Blocks vitamin K epoxide reductase, this is required for activation of Vitamin K preventing clotting factors 2,7,9,10 from being produced, and proteins C and S
3. Vit K medic applies gamma shaped bandage: vitamin K promotes gamma carboxylation of coagulation factors II, VII, IX, and X,C,S
4. Throm-beaver with II shaped teeth: factor II (thrombin)
5. Seven deadly sins: Factor VII
6. Nine lives cat: factor IX
7. FoX: factor X
8. GL: gamma carboxylation occurs at the glutamic acid residue on factor II, XII, IX, and X
9. V-KOR supply boat: vitamin K epoxide reductase (VKOR) converts vitamin K epoxide (inactive) into vitamin K (Active)
10. Corporal: vitamin K promotes gamma carboxylation of proteins C
11. Sergeant: vitamin K promotes gamma carboxylation of protein S
12. Corporal and sergeant hold their troops back: proteins C and S are anticoagulant factors
13. Remember: vitamin K contributes to coagulant and anticoagulant forces
14. Warhead destroying V-KOR supply ship: Warfarin inhibits vitamin K epoxide reductase (VKOR)
15. Incapacitated Vik K medic: inhibition of VKOR prevents activation of Vit K
16. Onset of action is not immediate, not for acute thrombotic events
17. Wounded VIII soldier: factor VIII is the first clotting factor to be reduced when starting warfarin
18. Delayed warhead detonation: warfarin onset of action is 8-12 hours, full clinical effect takes 3 days
19. Soldier with open mouth leaning on warfarin bomb: oral administration
20. Long tapering flag: Long half life
21. Para trooping soldier: Monitor warfarin using PT time, this is an extrinsic factor, like the paratrooper landing extrinsically
22. INtercom Radio worn by paratrooper: the international normalized ration (INR) is also used to measure warfarin activity
23. Goal INR 2-3 for prevention and treatment of thrombosis
24. Irregularly irregular heart signal: warfarin is used for long term anticoagulation in atrial fibrillation
25. Warfarin patrolling Iliofemoral river: used as DVT prophylaxis
26. Acute txt for DVT is IV heparin, warfarin is delayed onset
27. **Adverse effects**
28. Tarantula: warfarin can cross the placental barrier and can cause a hemorrhagic disorder or prevent carboxylation reactions in bone.
29. Soldiers charging past the injured corporal: the anticoagulation protein C is reduced early in warfarin therapy, resulting in a hypercoagulable state initially
30. Black soot on corporal: warfarin induced skin necrosis due to early hypercoagulable state
31. Heparin hunters patrolling the bridge: coadministration of heparin when starting warfarin therapy prevents the early hypercoagulable state (heparin bridge)
32. Skin necrosis risk is increased with a hereditary protein C deficiency
33. Distant Vit K medic reinforcements: warfarin anticoagulation can be reversed with vitamin K (delayed effect)
34. FFP fighter pilot: fresh frozen plasma (FFP) provides coagulation factors for immediate reversal of warfarin anticoagulation
35. CYP-450 chrome tank crushing warhead: warfarin is a substrate of cytochrome P-450 (increase P450 rifampin, phenobarbital, phenytoin, decrease effects) (decrease P450, antibiotics, antifungals, SSRI's, increase effects)
36.